

Coding & Solution

Date	15 March 2026
Team ID	
Project Name	Smart Lender:AI/ML-Based Loan Approval Prediction System

Coding & Solution

Solution Summary

Field	Details
Repository Link / URL	https://github.com/sowmya-123456/sowmya-123-AI-ML-Track-smartlender-Project-Template.git
Programming Language(s)	Python 3.8+, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost, Pandas, NumPy, Bootstrap
Key Features Implemented	• Loan approval prediction using Machine Learning • Comparison of Decision Tree, Random Forest, KNN, and XGBoost models • XGBoost selected as the best-performing model • User-friendly Flask web interface for loan prediction • Real-time prediction based on applicant details • Model serialization using Pickle • Responsive web interface with HTML/CSS
Pending / Incomplete Features	• User authentication and login system • Loan application history and dashboard • Database integration for storing applicant records • Email/SMS notification support • Cloud deployment enhancements
Setup / Run Instructions	1. Install Python 3.8+ and required libraries. 2. Clone or download the project. 3. Install dependencies using <code>pip install -r requirements.txt</code>. 4. Place the trained model file (<code>model.pkl</code>) in the project directory. 5. Run the application using <code>python app.py</code>. 6. Open a web browser and navigate to <code>http://127.0.0.1:5000</code> to access the Smart Lender application.

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The project follows a modular architecture by separating machine learning, model loading, Flask routes, and user interface components.
- Four classification algorithms—Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost—were trained and evaluated.
- XGBoost achieved the highest performance with **94.7% training accuracy** and **81.1% testing accuracy**, making it the final model used for deployment.
- The application provides fast, real-time loan approval predictions through a simple and responsive web interface.
- The project is designed to be scalable, allowing future integration with databases, authentication modules, cloud deployment, and advanced analytics.